

Retention, Value Segmentation & Churn Risk

Advanced MySQL project — window functions, CTEs, conditional aggregation, and self-joins, applied to 10,000 bank customers and ~626,000 transactions.



What this covers

Cohort retention — of customers who joined in a given month, how many are still transacting 1 / 6 / 12 months later, built with CTEs and conditional aggregation.

Customer value segmentation — quartile ranking (NTILE) across the whole customer base, plus a per-market leaderboard (RANK) by transaction volume.

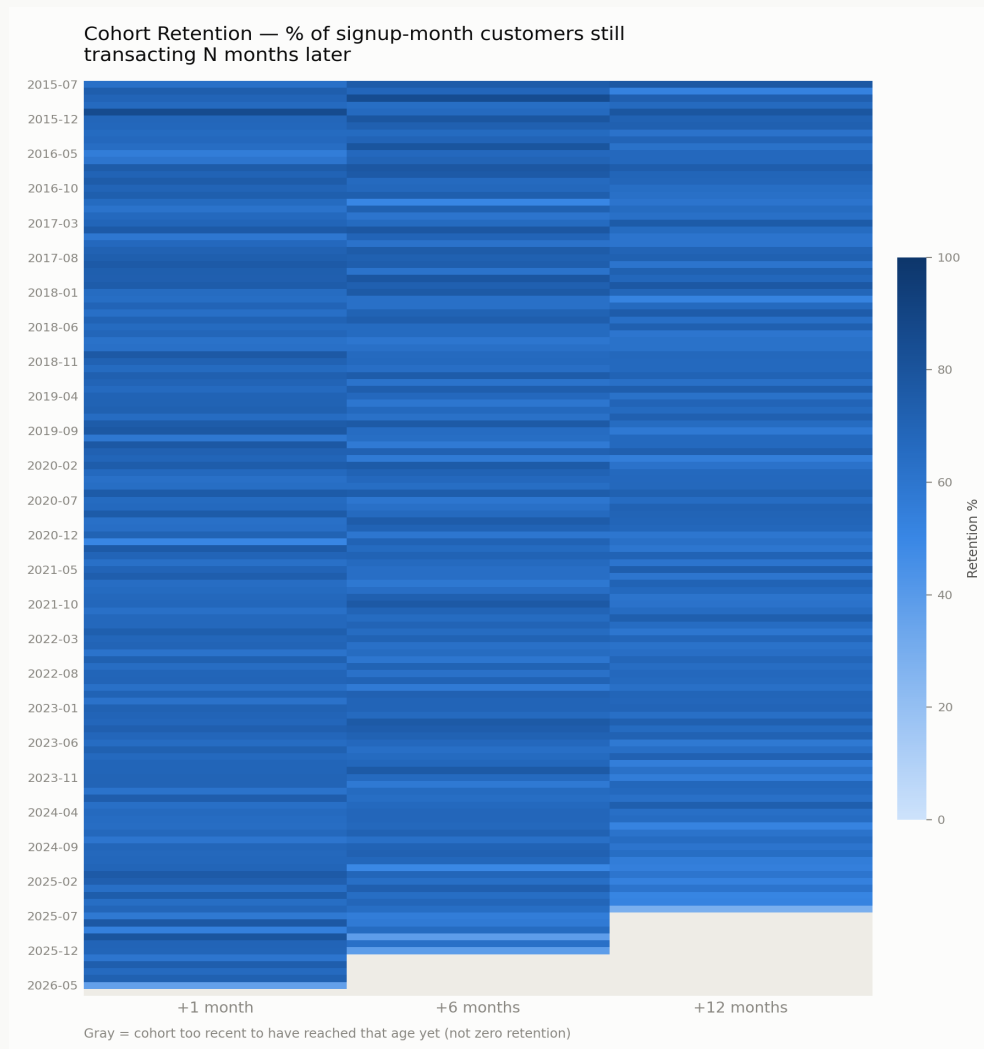
First-vs-last transaction & churn risk — self-joins to find each customer's earliest and latest activity, used to flag accounts going quiet before they're formally marked churned.

On the data: customer attributes (credit score, geography, tenure, balance, churn status) are the real Kaggle-style "Churn_Modelling" bank dataset. Transaction-level activity has no real equivalent in that source, so it was simulated per customer — grounded in each customer's real tenure and churn status, not random noise. Data extract date: **2026-06-30**.

COHORT RETENTION

Are newer customers sticking around better than older ones?

Each row groups customers by the month they signed up (a "cohort") and shows what share of that cohort was still transacting 1, 6, and 12 months after joining. Gray cells mean the cohort hasn't reached that age yet as of the data snapshot — not zero retention.



68%

AVG. RETENTION AT +1 MONTH

67%

AVG. RETENTION AT +6 MONTHS

65%

AVG. RETENTION AT +12 MONTHS

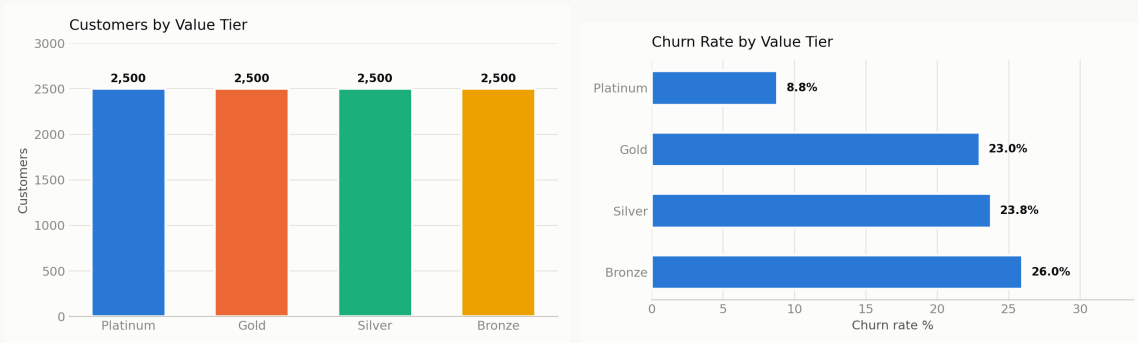
Retention (averaged across all cohorts old enough to have reached that milestone) drops fastest in the first month, then flattens — the customers still around at month 6 mostly make it to month 12. That first-month

drop is where a retention intervention (onboarding, a check-in nudge) would have the most leverage.

CUSTOMER VALUE SEGMENTATION

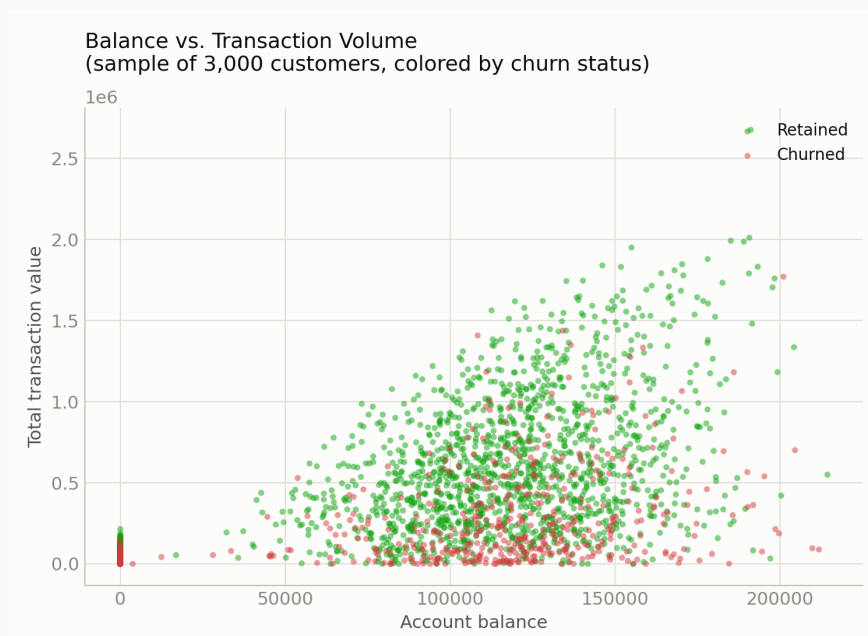
Who's actually valuable — and are we about to lose them?

Customers are split into four equal-sized quartiles (NTILE) by total transaction value, labeled Platinum through Bronze. This is a very different cut from account balance — a large idle balance and heavy day-to-day transaction activity are not the same thing.



Key finding: Platinum-tier customers churn at **8.8%**, while Bronze-tier customers churn at **26.0%** — roughly 3.0x higher. Value and retention move together here: the customers worth keeping most are, encouragingly, also the ones most likely to stay. It also means Bronze-tier churn is the bigger volume problem even though each individual account is worth less.

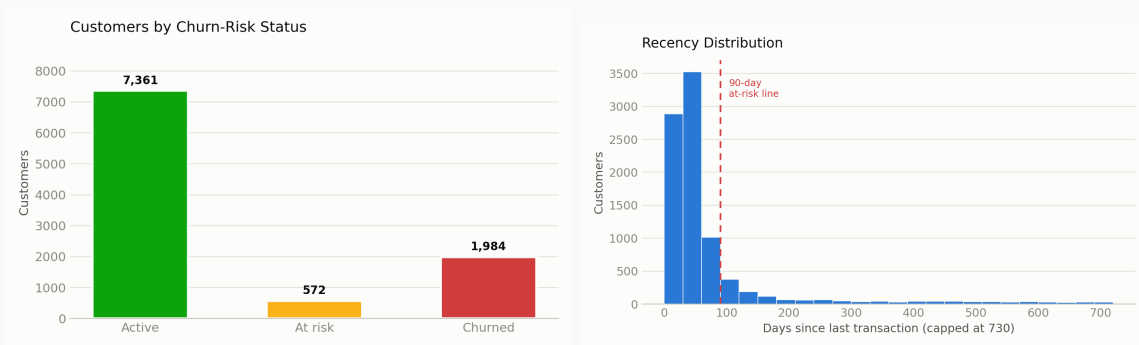
Balance vs. transaction volume



A customer's balance and how much they actually transact aren't the same signal — plenty of high-balance customers show light transaction activity. Churned customers (red) skew toward lower transaction volume across most balance levels, consistent with disengagement showing up in behavior before it shows up as a formal exit.

Who's going quiet before they're marked as churned?

Using a self-join to find each customer's earliest and latest transaction date (rather than a plain MIN/MAX aggregate), every customer is flagged Active, At risk (90+ days since their last transaction), or formally Churned.



Key finding: 572 customers (5.8%) are flagged At risk — still technically active accounts, but quiet for 90+ days. That's a meaningfully sized list that standard churn reporting (which only counts the 1,984 formally exited customers) misses entirely. It's the natural target list for a win-back campaign, before those accounts convert to formal churn.

Why a self-join instead of MIN()/MAX()

The self-join approach (join the transactions table to itself, keep only rows with no earlier/later match) is the classic interview version of this problem. In production, the MIN()/MAX() + GROUP BY version is what you'd actually ship — same result, far better performance at scale, since the self-join is roughly $O(n^2)$ without a very targeted index. Both versions are in the project's query file; this project validated they return identical results before treating either as final.

SUMMARY

Three takeaways

1. Retention is won or lost in month one.

Across mature cohorts, the steepest drop in transacting customers happens between signup and month one; cohorts that survive to month six mostly carry through to month twelve. Onboarding and first-30-days engagement are where a retention effort would pay off fastest.

2. High-value customers are already the most loyal — the risk is concentrated lower down.

Platinum-tier churn (8.8%) is well below Bronze-tier churn (26.0%). The bank isn't at risk of losing its best customers disproportionately — the churn volume problem is concentrated in the lower-value tiers, which changes where a retention budget should go.

3. There's a meaningful gap between "active" and "actually engaged."

572 customers (5.8%) haven't transacted in 90+ days without being formally marked as churned. Any churn metric based only on the formal exit flag understates the real at-risk population by a meaningful margin.

Full project — schema, data generation, all query files (with both the self-join and aggregate versions), and a step-by-step execution runbook — available alongside this report.